



A Review of Soil Thermal Conductivity Measurement Techniques, Challenges, and Future Directions

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Abstract

Soil thermal conductivity (λ) plays a critical role in regulating heat dynamics at the land surface. In agricultural soils, it directly affects crop performance and can also influence soil management strategies. Nowadays, understanding the significance of soil λ has become vital for promoting sustainable agricultural practices. This review explores a wide range of measurement techniques for soil λ , including both traditional and advanced methods, and focuses on their applications, benefits, and limitations. Traditional steady-state methods—such as the guarded hot plate, thermal cell, steady-state needle probe, and box probe, are compared with transient techniques, including time-domain reflectometry, laser flash, transient hot wire, transient plane source, heat pulse, and transient line heat source. Although steady-state methods provide high accuracy in controlled laboratory conditions, they usually require extended testing times and are influenced by variations in soil water content and texture. Transient approaches, while more rapid and suitable for field conditions, find difficulties due to soil heterogeneity and environmental variability. Emerging technologies, such as fiber optic sensors and wireless sensor networks, give non-invasive, real-time monitoring with minimal soil disturbances, however, they are limited by higher prices and technological complexity. We evaluated them for their accuracy, efficiency, and adaptability to diverse agricultural conditions. This review underscores the necessity of precise soil λ measurements to enhance soil management strategies, improve resilience to environmental changes, and support sustainable agriculture. Important challenges in methods include calibration, sensitivity to soil characteristics, and practical field usability. It provides a valuable resource for researchers and practitioners, offering insights into integrating soil λ measurements into modern agricultural systems. Finally, future research directions are proposed, focusing on leveraging wireless sensor network technologies and machine learning to deepen our understanding of soil heat dynamics and their implications for sustainable farming practices.

1 Introduction

A critical factor influencing the heat dynamics of agricultural soils is the soil thermal conductivity, λ [1, 2]. Soil λ affects key processes such as seed germination [3, 4], root development [5], and crop productivity [6]. Understanding the influence of soil properties, especially λ , on agricultural operations has become increasingly crucial for enhancing agricultural resilience and ensuring food security [7–12]. Soil λ generally ranges from approximately 2 to 8 W m⁻¹ K⁻¹ for mineral soil [13], depending on factors such as soil texture, moisture content, and mineral composition [13–15]. In contrast, organic soils typically exhibit much lower values, ranging between 0.1 and 0.5 W m⁻¹ K⁻¹, even under moist conditions [16]. This reduction in λ is primarily attributed to their higher porosity and lower bulk density, which limit heat transfer efficiency [15, 17].

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This review aims to explore the techniques used to measure soil λ and evaluate its applications, limitations, and adaptability in agricultural contexts. Accurate λ measurements are essential for effective soil management and crop yield optimization across diverse environmental conditions. For instance, knowledge of soil λ supports informed decision-making regarding irrigation practices [3, 18], soil amendments [6, 19–21], and fertility management [22–24], all of which are vital for maintaining soil health and productivity.

Numerous methods exist for measuring soil λ , each with distinct advantages and limitations. Conventional approaches, such as steady-state methods and transient techniques, have been widely employed in laboratory and field studies [2, 25–28]. Recent technological advancements, including the development of fiber optic sensors [29] and wireless sensor networks [25], have introduced innovative opportunities for real-time data collection and monitoring. Beyond individual crop performance, soil λ significantly affects soil water retention [30], texture [2], water content (μ) [31], structure [32], temperature [33], organic content [9], tillage practices [6, 34], irrigation methods [11], and vegetation cover [35, 36].

2 Methodology

This review adopts a systematic and comprehensive methodological approach to evaluate the body of literature on soil λ . A detailed investigation was conducted, focusing on publications from 2015 to 2026, using the Web of Science and Scopus databases, both renowned for their extensive collections of peer-reviewed publications across disciplines such as soil science and engineering. The search strategy utilised a combination of specific keywords and phrases to ensure an exhaustive assessment of relevant studies. Key terms included “Soil Thermal Conductivity”, “Soil Thermal Measurement Techniques”, “Traditional Steady-State Methods”, “Transient Techniques”, and “Advanced Methods”. This targeted approach facilitated the identification of high-impact research articles addressing the intricate relationships between soil λ techniques, soil properties, and agricultural systems.

The research process involved an in-depth review of 147 academic publications to determine their relevance and contributions to the field. Each selected manuscript underwent a rigorous critique to extract key findings, methodological details, and implications specific to soil λ in agricultural contexts. Additionally, the review highlighted significant advancements in soil science and agriculture, focusing on innovative measurement methods and technological

developments that enhance understanding of soil thermal dynamics.

This review also examines critical aspects of soil λ measurement techniques, including measurement processes and procedures; sensitivity to variations in soil properties; advantages and limitations of different methods; equipment and tools employed; and methodological constraints and challenges.

3 How Can Thermal Conductivity (λ) of Measurements in Agr-Environmental Assessments?

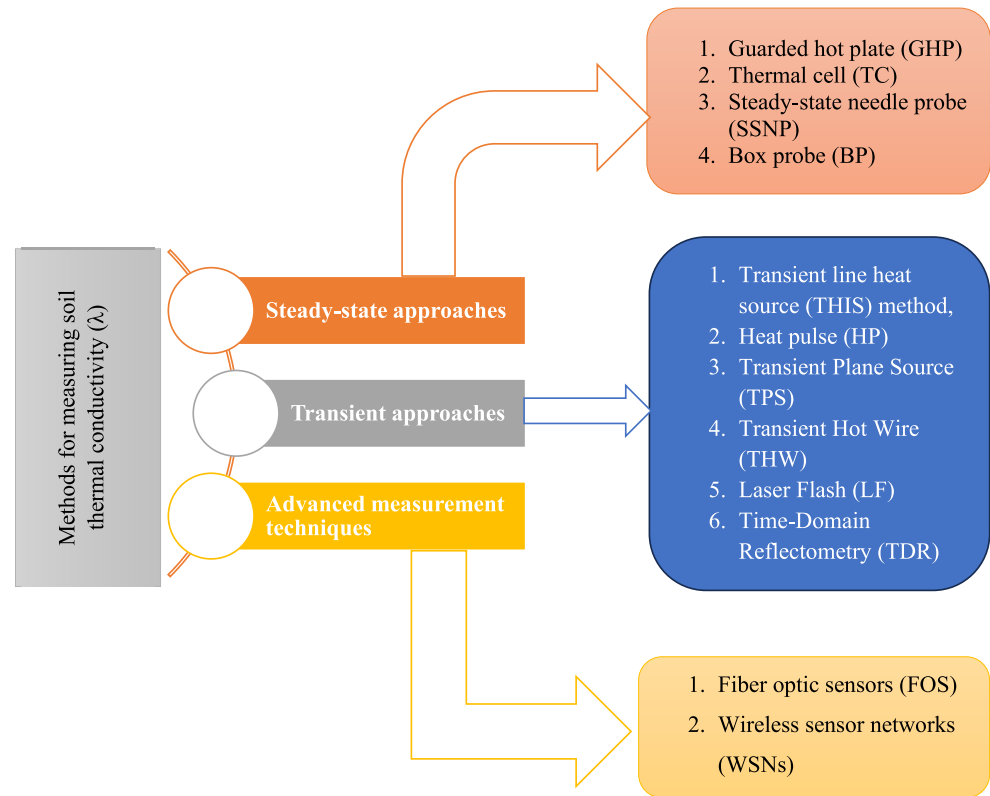
Determining soil λ is crucial for various engineering and environmental applications, which directly impacts agricultural productivity, soil health, and environmental sustainability. Various techniques have been developed to measure soil λ , broadly categorized into steady-state and transient methods (Fig. 1) [37–39]. In addition to traditional strategies, several novel methods have been introduced to improve measurement accuracy and efficiency, integrating advanced technologies and data-driven approaches. The primary techniques are outlined in the next sections.

3.1 Steady-State Methods of Soil λ Measurements

Steady-state methods of measuring soil λ are crucial for understanding heat transmission in soils, particularly in agricultural and environmental applications. These methods involve maintaining a uniform temperature gradient and measuring the heat flow through a soil sample once thermal equilibrium is reached. Several steady-state techniques, including the guarded hot plate, thermal cell, steady-state needle probe, and box probe methods, are widely used for laboratory measurements (Fig. 1) [40, 41]. While these approaches provide high accuracy and reliability, they typically require longer testing durations compared to transient methods. Below are the key steady-state techniques for assessing soil λ .

3.1.1 Guarded Hot Plate (GHP) Method

The GHP method is a widely recognized and reliable technique for measuring soil λ under steady-state conditions. Its ability to maintain a uniform temperature gradient across a soil specimen ensures high measurement accuracy, making it a valuable tool for assessing thermal properties. This method operates by sandwiching the soil specimen between a hot plate and a cold plate, creating a controlled temperature difference (Appendix: Fig. 1). Once steady-state conditions are achieved, the heat flux is measured, allowing for

Fig. 1 Soil thermal conductivity measurement methods

precise calculations of λ . To minimize heat loss and maintain stable environmental conditions, the entire apparatus is often enclosed in an insulated chamber, which helps ensure consistent and accurate results [42, 43].

Recent advancements in GHP design have significantly improved its precision and efficiency. The integration of heat flux meters allows for more accurate heat flow measurements, while enhanced heater designs ensure uniform temperature distribution across the sample. Furthermore, the use of advanced insulation materials has helped minimize external heat loss, leading to more reliable thermal conductivity assessments thereby [44]. Validation studies indicate that GHP measurements closely align with established reference materials, with deviations typically within 2–3% of expected values. However, thermal resistance at the contact points between the specimen and the plates can introduce minor inaccuracies, particularly under low-pressure or low-temperature conditions. Nevertheless, under standardized testing conditions, these effects remain minimal, ensuring dependable λ evaluations (Appendix: Fig. 1).

Despite its high accuracy, the GHP method presents several challenges and limitations that must be addressed to optimize its application. One of the primary concerns is heat loss through the apparatus, which can lead to an underestimation of λ if insulation is inadequate. The extent of this heat leakage is influenced by the width and material composition of the thermal barriers, making proper insulation a

crucial factor in maintaining measurement accuracy. Additionally, sample size can impact the reliability of results, as larger samples may introduce inhomogeneity, affecting temperature distribution, while smaller samples may fail to accurately represent field conditions. Soil properties, such as texture, porosity (Φ), and water content (μ), also play a significant role in determining λ , with variations in organic matter and mineral composition influencing heat transfer dynamics. Maintaining precise temperature control throughout the testing process poses another challenge, as fluctuations in ambient conditions can prolong testing durations and introduce errors in measurement. The complexity of integrating multiple interacting elements, such as Proportional-Integral-Derivative (PID) controllers for temperature regulation, further adds to the difficulty of achieving stable conditions during testing.

To enhance the accuracy and reliability of the GHP method (Table 1), future research should focus on developing advanced insulation materials and refining temperature control mechanisms. Standardizing measurement procedures will help improve consistency across studies, while the integration of automated control systems for real-time temperature adjustments can enhance precision. Moreover, validating laboratory findings with field measurements is essential to ensure that GHP assessments accurately reflect real-world soil conditions. Addressing these limitations will allow the GHP method to continue serving as a gold

Table 1 Research gaps and future work of traditional steady-state methods for measuring soil λ

Method ID	Problems	Description of research gaps	Future work
Table 1a			
a) Guarded Hot Plate (GHP)	Heat loss	GHP setups often experience heat loss, leading to underestimated λ	Improve insulation materials and compare traditional vs. advanced systems to minimize heat loss
	Sample size	Effects of varying sample sizes on λ are unclear	Standardize sample sizes for different soils and analyze their influence on λ accuracy
	Moisture content variability, temperature control improvements	Inconsistent μ and unstable temperature affect λ accuracy	Develop protocols for μ control and advanced PID temperature systems to stabilize GHP measurements
	Validation with field measurements	Few comparisons exist between lab and field GHP results	Conduct validation studies to align laboratory and field λ data
	Impact of soil composition	Influence of OM and minerals on λ not fully understood	Study how different soil constituents affect λ to improve soil behavior prediction
Table 1b			
b) Thermal Cell (TC) method	Soil heterogeneity and scale	TC often applied to homogeneous soils, limiting field relevance	Adapt TC protocols to account for layered and variable soils
	Moisture dynamics	μ variability effects on λ under transient conditions are underexplored	Study transient μ impacts and develop control techniques for dynamic environments
	Temperature range effects	Narrow temperature ranges limit λ understanding in diverse climates	Expand tests across wider temperature intervals and mulching systems
	Equipment and calibration improvements	Current TC setups show low sensitivity and high ambient interference	Design durable TC systems and improve calibration for greater accuracy
	Effect of Soil chemicals	Effects of OM and fertilizers on λ remain unclear	Examine how different amendments influence λ in agricultural soils
	Integration with remote sensing	λ data rarely linked with remote sensing for large-scale mapping	Combine TC results with RST to model and map λ regionally
Table 1c			
c) Steady-State Needle Probe (SSNP)	Limited soil textures and conditions	Research mostly targets few soil textures, reducing applicability	Test across varied soil types and μ to enhance reliability and generalization
	Effect of soil structure and composition	Compaction and layering effects on λ often ignored	Quantify structure impacts using imaging and statistical modeling
	Integrating with different λ measurement techniques	Studies rarely combine SSNP with other λ techniques	Integrate SSNP with transient and heat-flux methods for validation
	Application in climate change studies and temporal variability	Limited exploration of λ under changing climate or diurnal cycles	Investigate λ dynamics over time for improved climate models
	Calibration methods and data analysis	Lack of standard calibration and data interpretation protocols	Standardize SSNP calibration and apply ML/statistical tools for data analysis
Table 1d			
d) Box Probe (BP) method	Soil heterogeneity and composition effects	The influence of soil heterogeneity, OM, and minerals on λ is not fully understood	Investigate how variations in soil texture, OM, and mineral content affect λ accuracy under diverse environmental conditions
	Variability in moisture content (μ)	Changes in μ significantly alter λ , yet the μ - λ relationship across soil types remains unclear	Conduct studies to quantify μ - λ interactions and develop techniques to control μ during λ measurement
	Temperature range limitations	Most BP studies are restricted to narrow temperature ranges	Extend testing to wider temperature intervals to enhance BP applicability in varied climatic conditions
	Calibration protocols, and Integration with advanced technologies	Lack of standardized calibration and limited integration with modern sensing systems	Develop universal calibration standards and explore advanced technologies for data acquisition and analysis
	Field application challenges	Current research mainly focuses on lab-based λ measurements, limiting understanding of in situ behaviour	Design portable, wireless BP prototypes to enable accurate field measurements under real soil conditions

standard for soil λ evaluations, providing valuable insights for agricultural management, engineering applications, and climate research.

3.1.2 Thermal Cell (TC) Method

The TC method is a widely used technique for measuring soil λ , employing various devices and approaches specifically designed to assess soil heat transfer properties with high accuracy [45, 46]. This method is crucial in scientific research and agricultural applications [47], where a precise understanding of soil λ is necessary for optimizing soil management and predicting thermal behaviour [47].

Typically, the TC method involves placing soil in a thermal cell equipped with a heat source and temperature sensors to monitor heat transfer dynamics [47–49]. This setup enables researchers to measure steady-state heat flux, providing reliable and consistent results under controlled conditions. One of the key advantages of this approach is its ability to facilitate in-situ measurements, allowing data collection that closely reflects natural soil conditions without requiring extensive sampling [50]. Additionally, TC methods are particularly effective in maintaining stable temperature gradients, leading to precise evaluations of λ in different soil types [47, 48].

Despite these benefits, several limitations must be considered when using the TC method [15]. The equipment and experimental setup required for TC measurements can be expensive and complex, potentially restricting access for some researchers and agricultural practitioners [51]. Furthermore, the insertion of the TC apparatus into the soil can disturb its structure, leading to potential measurement inaccuracies [48]. Another critical limitation is the sensitivity of TC to temperature gradients, which can introduce errors if not properly managed, necessitating frequent calibration to maintain accuracy [27]. Additionally, measuring soil λ at greater depths remains a challenge, often requiring complementary methods for a more comprehensive thermal analysis (Appendix: Fig. 2).

Despite these challenges, Table 1b highlights the specific limitations of the TC method, particularly regarding soil heterogeneity, water content and temperature variations, which can all affect the accuracy of λ measurements [48]. Addressing these challenges requires advancements in sensor technology, calibration procedures, and data processing methodologies to improve measurement reliability. Additionally, integrating soil chemical analysis and remote sensing technologies into TC measurements could further enhance precision and applicability [27].

The versatility of the TC method remains a notable strength, as it can be applied across various soil textures and enables real-time data collection, making it a valuable tool

for understanding soil thermal dynamics [49]. However, factors such as μ levels, compositional properties, and temperature fluctuations can significantly influence λ results, emphasizing the importance of selecting the appropriate methodology and instrumentation for specific research conditions.

Technological advancements continue to enhance the accuracy and efficiency of TC measurements, making them increasingly viable for diverse applications in soil science and agriculture. However, persistent challenges related to high costs, potential soil disturbance, and depth limitations underscore the need for continued refinement of these methods. Future research should focus on improving the precision, adaptability, and scalability of the TC technique through advanced measurement systems, better environmental controls, and integration with complementary analytical tools, as outlined in Table 1b. These improvements will help solidify the role of TC in soil thermal studies, ensuring its continued relevance in agricultural and environmental research.

3.1.3 Steady-State Needle Probe (SSNP) Method

The SSNP method is a widely utilized technique for measuring soil λ , particularly in both laboratory and field settings [27, 52]. This method involves inserting a needle-like thermal probe into the soil, where it applies a constant heat source. As heat propagates through the soil, a temperature gradient stabilizes, allowing for the determination of λ based on the thermal flow and corresponding temperature changes over time [49, 53].

One of the primary advantages of the SSNP method is its ability to provide accurate measurements in unsaturated soils, as it accounts for the dynamic state of pore water content and p_b [54] (Appendix: Fig. 3). Factors such as p_b , μ , and pore size distribution significantly influence the measured λ values [49]. Also, soil composition, including the presence of organic components and variations in particle sizes, can impact λ results. The method's sensitivity to these factors can lead to variability in measurements, particularly in sand soils with fluctuating μ and p_b levels. Despite its relatively simple setup, requiring minimal equipment and preparation, its effectiveness is partially dependent on the geometry and insertion accuracy of the probe [55].

However, certain limitations must be acknowledged. Poor thermal contact between the probe and the soil, often caused by insertion damage, can result in underestimations of λ [56]. Additionally, the mechanical fragility of thin probes makes them susceptible to bending, further affecting measurement accuracy. Furthermore, SSNP assumes that thermal measurements achieve a logarithmic regime, but in practice, this condition is not always met, potentially

leading to significant errors in λ estimations, particularly in materials like snow (Appendix: Fig. 3). Thorough calibration is essential to ensure reliable λ results [55, 57].

Despite these challenges, the SSNP method remains highly effective for agri-environmental studies and engineering applications, where understanding soil thermal behaviour is crucial. It has practical implications for optimizing irrigation strategies, assessing geothermal energy potential, and evaluating soil performance in construction projects. By providing precise λ measurements, SSNP contributes valuable insights into soil heat dynamics, supporting better management and resource utilization. However, the selection of the appropriate technique depends on specific soil properties and environmental factors during measurement.

Future improvements in probe design and data analysis methodologies could help mitigate some of the existing limitations. Table 1c systematically outlines the research gaps and future directions for SSNP, highlighting issues related to the limited range of soil textures and conditions examined, the influence of soil morphology and composition, and the need for better integration with other λ measurement techniques. Furthermore, it underscores the necessity for enhanced calibration protocols, improved data processing techniques, and expanded applications in climate change research. Future research should focus on broadening the range of tested soil types and conditions, integrating SSNP with complementary measurement methods, and refining calibration and analytical approaches to increase accuracy and applicability across diverse soil environments.

3.1.4 Box Probe (BP) Method

The Box Probe (BP) method, used to measure λ [53], is particularly suitable for small soil samples and is designed to provide accurate λ measurements while requiring minimal equipment, making it a cost-effective and practical alternative to more complex techniques [58, 59]. The BP method operates by utilizing a box-like apparatus to assess the effective λ of soil samples through controlled heat transfer (Appendix: Fig. 4). Recent modifications have reduced the number of required sensors, thereby lowering costs and simplifying implementation. However, there are still research gaps concerning its accuracy, applicability, and efficiency under varying soil conditions and environmental settings. The BP method consists of several key components, including a rectangular probe unit that houses a heating element and embedded temperature sensors, ensuring uniform heat distribution within the soil (Appendix: Fig. 4). A thermal insulator housing minimizes heat loss to the surroundings, maintaining stable temperature readings. The heating element, typically an electric resistance heater, supplies controlled heat to the soil, while thermocouples monitor

temperature changes over time. A data acquisition system records and logs temperature variations, while a power supply energizes the heating element and sensors [53]. These components work together to ensure consistent and reliable λ measurements, but certain limitations remain regarding their effectiveness in varied soil environments.

Several factors influence the accuracy of λ measurements obtained using the BP Method (Appendix: Fig. 4). Soil μ plays a significant role, as higher μ levels generally increase λ due to enhanced heat transfer efficiency [60]. Additionally, soil composition, including mineral content and particle size, affects thermal properties, with finer particles typically exhibiting higher λ values [59]. Similar to Appendix: Fig. 4, the temperature gradient or the rate at which temperature changes, can also impact readings, particularly if the gradient is too steep. Soil structure is another critical aspect, as compacted or aggregated soils may exhibit different thermal properties compared to loose, granular soils [61, 62]. Finally, ambient temperature conditions can introduce measurement inaccuracies, as external temperature fluctuations influence heat loss from the probe. Despite these challenges, the BP method has demonstrated good agreement with reference techniques, such as the hot disk method, reinforcing its reliability in measuring soil λ [58]. Its simplified setup makes it particularly useful in field conditions, where traditional methods might be impractical or cumbersome. However, while the BP method offers several advantages, its suitability depends on specific application requirements. For instance, in cases requiring multi-directional heat transfer analysis or where soil properties vary significantly, alternative methods such as the multi-directional apparatus or the heated probe method may be more appropriate. The selection of a measurement technique should consider available resources, required accuracy, and environmental factors at the study site.

Future research should address the existing gaps and explore ways to enhance the BP method's effectiveness. Table 1d systematically outlines the limitations of the BP method, particularly in terms of soil heterogeneity, water content variability, and temperature range restrictions. It also emphasizes the necessity for better calibration protocols, integration with advanced technologies, and improvements in field applications. Moving forward, research should focus on understanding how soil composition and moisture influence λ , expanding the temperature range examined, and developing standardized calibration procedures to increase the accuracy and applicability of the BP method in both laboratory and field settings.

3.2 Transient Methods of Soil λ Measurement

Transient methods are widely utilized for measuring soil λ due to their efficiency and capability in delivering accurate λ assessments under varied conditions. Unlike steady-state techniques, which require prolonged measurement periods, transient methods capture rapid temperature fluctuations over time, enabling real-time and in situ analysis of soil thermal behavior. These techniques are particularly valuable for applications in agriculture, geotechnical engineering, and environmental monitoring, where understanding dynamic heat transfer is essential for optimizing irrigation strategies, improving construction safety, and assessing climate-related impacts.

By responding to instantaneous changes in thermal conditions, transient methods provide a more flexible and time-efficient approach to soil λ measurement. They allow for the determination of λ in heterogeneous soil structures with minimal disturbance to soil composition. Some of the most widely employed transient techniques include the Transient Line Heat Source (THIS) method, Heat pulse (HP), Transient Plane Source (TPS), Transient Hot Wire (THW), Laser Flash (LF), and Time-Domain Reflectometry (TDR), each characterized by distinct operational principles and applications scopes (Fig. 1) [38, 40]. The THIS method involves a continuous heat source with a line, which allows for measuring soil heat properties by recording the temperature with time, while the HP method utilizes a discrete pulse of heat applied at a single point, measuring change to infer heat conductivity (Abed Gatea [63; 64]). These techniques offer significant advantages in terms of speed, adaptability, and precision, making them indispensable for soil thermal analysis across a broad spectrum of scientific and industrial fields.

3.2.1 Transient Line Heat Source (THIS) Method

The Transient line heat source (THIS) method is a well-established and widely employed technique for measuring soil λ in experimental settings [1]. The method operates by inserting a needle-like thermal probe into the soil, where a constant heat pulse is applied [65]. The temperature increase in the surrounding soil is recorded, and λ is determined based on the rate of temperature rise [66].

One of the major advantages of the THIS method is its ability to deliver rapid and accurate λ measurements without requiring extensive soil sampling [27, 67, 68] as shown in Appendix: Fig. 5. This makes it particularly suitable for in-field measurements, where soil properties can be assessed without disrupting the natural structure. However, as depicted in Appendix: Fig. 5, certain factors can negatively influence measurement accuracy, including soil water

content (μ) variability and inhomogeneity [48, 69, 70]. Furthermore, the physical setup, particularly the contact between the probe and the soil, as well as the measurement duration, can introduce errors if not carefully managed [27, 65, 68].

The THIS method has been applied in various studies to assess soil thermal properties across different groundwater depths, demonstrating its practical relevance in field applications [71]. It has proven especially beneficial in agriculture, where accurate λ measurements assist in optimizing irrigation strategies and understanding crop growth thermal dynamics. Additionally, in environmental science, the method plays a critical role in evaluating the impact of climate change on soil ecosystems by characterizing thermal properties. By facilitating efficient soil λ monitoring, the THIS method contributes to sustainable soil management, reinforcing its significance in both research and practical applications [72].

Despite its advantages, the THIS method presents several limitations, as outlined in Table 2a. Key challenges include issues related to calibration, data analysis, and field applications, highlighting the need for enhanced integration of soil property considerations and comparative analyses with other methods. Future research should focus on refining calibration techniques, developing advanced data analysis models, and conducting long-term field studies to improve the accuracy, reliability, and applicability of the THIS method across diverse environmental conditions (Table 2a).

3.2.2 Heat Pulse (HP) Method

The Heat Pulse (HP) method is a widely used technique for assessing soil λ , which is crucial for understanding heat and mass transfer processes in soils. The HP can be classified into single and dual-probe heat pulse techniques (SPHP, and DPHP, respectively). The SPHP method uses the fibre optic cable for both the thermal application and temperature detection, while DPHP systems sense the temperature change at a specific distance from the heat pulse source [73, 74]. The HP method operates by applying a heat pulse to the soil and measuring the resulting temperature changes to determine thermal properties [75, 76]. This method is particularly significant due to its ability to provide accurate and continuous measurements in both laboratory and field environments (Abed Gatea [63; 77]).

The measurement process of the HP method involves critical steps to ensure an accurate assessment of soil λ . Initially, a heat pulse is introduced into the soil using a specialized probe, which consists of a heater and temperature sensors [75] (Appendix: Fig. 6). The temperature response is subsequently monitored over time, allowing researchers to calculate the soil's thermal conductivity based on

Table 2 Research gaps and future work for the transient techniques for measuring soil λ

Methods ID	Problems	Description of research gaps	Future work
Table 1a			
Transient Line Heat Source (THIS) method	Improved calibration techniques, automated measurement systems	Existing calibration procedures do not fully account for soil variability and require manual operation	Establish standardized calibration protocols considering soil texture, structure, and μ ; develop automated systems for real-time λ data collection and analysis
	Enhanced data analysis	Current models often oversimplify transient heat transfer, especially in heterogeneous soils	Create advanced numerical models incorporating soil variability and heat transfer dynamics; validate experimentally across soil types and compare with other λ methods
	Integration of soil properties	Limited inclusion of soil physical parameters, such as density and porosity, in λ evaluation	Investigate relationships between λ and soil physical properties to develop predictive models and benchmark against traditional methods for accuracy and reliability
	Field application challenges, long-term monitoring	Field results vary with environmental conditions; short-term measurements miss seasonal effects	Conduct field trials in diverse climates; implement long-term monitoring to track λ variation with μ and temperature changes
	Comparative studies	Few studies directly compare THIS with other λ measurement techniques	Perform comparative assessments to evaluate THIS performance in terms of accuracy, repeatability, and usability
Table 1b			
Heat pulse (HP) method	Calibration	Current HP calibration procedures are not universally applicable across soil types, causing inaccuracies	Establish standardized calibration protocols considering texture and μ ; apply machine learning to develop adaptive calibration models and compare with traditional HP methods
	Advanced data analysis techniques	Present HP analyses oversimplify heat transfer in heterogeneous soils	Develop advanced models incorporating soil variability, transient flow, and 3D effects to improve λ estimation accuracy
	Integration with soil property measurements	Limited linkage between λ and other soil attributes such as porosity and density reduces understanding of soil thermal behaviour	Investigate correlations between λ and physical–chemical soil traits to build comprehensive HP-based predictive frameworks
	Field application studies	Few long-term or large-scale studies evaluate HP performance under real environmental variability	Conduct multi-season field experiments to assess HP robustness and reliability across diverse soils and climates
	Impact of environmental factors	External factors (e.g., vegetation, surface temperature) affecting λ accuracy remain underexplored	Quantify the impact of vegetation and temperature using combined HP and remote-sensing approaches to enhance real-world applicability
	Automated measurement systems	Manual operation restricts scalability and real-time monitoring	Integrate HP with IoT sensors and machine-learning algorithms for automated data collection and analysis; validate performance in field conditions
Table 2c			

Table 2 (continued)

Methods ID	Problems	Description of research gaps	Future work
d) Transient Hot Wire (THW) method	Soil heterogeneity	The influence of soil texture, organic matter, and mineral composition on λ measured by THW is not fully understood	Investigate how soil structure and components affect λ accuracy across different environmental conditions
	Soil μ and temperature range variability	Relationships between μ variation, temperature range, and λ are insufficiently characterized	Assess how changes in μ and temperature affect λ and expand THW testing to broader climatic conditions
	Calibration procedures	Lack of standardized calibration protocols reduces measurement reliability	Develop and validate consistent calibration frameworks to ensure comparability across laboratories and field applications
	Field application challenges	Limited data on THW performance in situ; soil disturbance and variability complicate field use	Conduct field trials using hybrid THW setups; evaluate performance and accuracy under real environmental conditions
	Integration of advanced technologies	Current THW models lack integration with modern sensors and predictive systems	Combine THW with advanced sensing and modeling techniques to enhance data precision and enable predictive λ estimation
Table 1c			
c) Laser Flash (LF) method	Sample heterogeneity	The effects of soil composition, texture, and organic/mineral content on λ measured by LF remain insufficiently explored	Investigate how soil heterogeneity influences λ under different conditions using imaging and analytical tools to correlate composition with LF-based results
	Calibration standards	Absence of unified calibration protocols limits comparability across studies	Develop standardized calibration frameworks validated across laboratories; apply statistical and machine learning tools to optimize calibration accuracy
	The μ effects	The influence of μ on LF λ measurements has not been systematically examined	Study λ variation across μ levels in diverse soils; design control and predictive models to enhance accuracy in varying moisture conditions
	Field application challenges	LF is primarily tested in controlled settings, with limited field application data	Develop and test portable LF systems; integrate with remote sensing to enable hybrid field–lab measurement approaches
	Integration with advanced technologies	Limited implementation of modern sensors and automated systems in LF setups	Integrate LF with advanced sensing and data acquisition technologies to improve precision and operational efficiency
Table 1d			
d) Laser Flash (LF) method	Sample heterogeneity	The effects of soil composition, texture, and organic/mineral content on λ measured by LF remain insufficiently explored	Investigate how soil heterogeneity influences λ under different conditions using imaging and analytical tools to correlate composition with LF-based results
	Calibration standards	Absence of unified calibration protocols limits comparability across studies	Develop standardized calibration frameworks validated across laboratories; apply statistical and machine learning tools to optimize calibration accuracy
	The μ effects	The influence of μ on LF λ measurements has not been systematically examined	Study λ variation across μ levels in diverse soils; design control and predictive models to enhance accuracy in varying moisture conditions
	Field application challenges	LF is primarily tested in controlled settings, with limited field application data	Develop and test portable LF systems; integrate with remote sensing to enable hybrid field–lab measurement approaches
	Integration with advanced technologies	Limited implementation of modern sensors and automated systems in LF setups	Integrate LF with advanced sensing and data acquisition technologies to improve precision and operational efficiency
Table 1f			

Table 2 (continued)

Methods ID	Problems	Description of research gaps	Future work
f) Time-Domain Reflectometry (TDR) method	Soil composition effects	The influence of organic matter and mineral composition on λ measured by TDR remains poorly understood	Investigate how soil constituents affect λ accuracy using machine learning to model and predict composition-based variability
	The μ and temperature effect	The relationship between μ and λ across varying moisture and temperature conditions is underexplored	Study λ variations under different μ and temperature ranges; develop standardized moisture-control and temperature calibration protocols
	Calibration protocols	Absence of standardized calibration methods limits reproducibility of λ measurements	Develop and validate consistent calibration procedures using machine learning models; test and compare across laboratories and field settings
	Field application challenges	TDR field applications face issues of soil disturbance and environmental variability	Design adaptive sensor networks for in situ λ measurements; conduct field trials to assess accuracy and reliability
	Long-term monitoring	Limited research on the feasibility of continuous λ monitoring using TDR	Develop IoT-based TDR systems for long-term soil λ monitoring; validate performance through extended field experiments

the rate of heat dissipation (Abed Gatea [63; 78; 79; 80]). This process is based on the principles of transient heat flow, where the speed and pattern of temperature change provide insights into the soil's thermal properties [81]. The effectiveness of this method is enhanced by its capacity to capture variations in λ , offering a detailed understanding of soil conditions across different environmental scenarios [82]. The accuracy of the HP method is significantly influenced by several soil properties, such as μ and soil texture [76]. Soil water content plays a crucial role in determining the thermal behaviour of soils, as higher μ levels generally increase λ due to the enhanced heat transfer facilitated by water present within the soil matrix [83]. Similarly, soil texture impacts λ , as different soil types, ranging from sandy to clayey compositions, exhibit distinct thermal properties that affect heat pulse readings [82]. These variations necessitate careful calibration and interpretation of data to ensure precise assessments, highlighting the importance of understanding soil-specific characteristics when employing the HP method in diverse environmental settings. The HP method offers several advantages that enhance its utility in soil λ assessments (Appendix: Fig. 6). One of the primary benefits is its precision in measuring thermal properties, which is crucial for accurate soil analysis [83]. Additionally, the non-destructive nature of the method allows for repeated measurements without altering the soil's natural state, thereby preserving the integrity of the study area [82]. Furthermore, the HP method provides rapid results, which is particularly beneficial in dynamic environments where timely data is essential for decision-making processes [78]. Its effectiveness across various soil conditions, including different μ levels and textures, further underscores its versatility and adaptability in field applications, making it a preferred choice for soil scientists and researchers working in diverse environmental settings (Abed Gatea [63]).

Several case studies have demonstrated the effectiveness of the HP method in assessing soil λ . For instance, research conducted on sandy and peat soils highlighted the method's capability to accurately measure thermal properties under varying water content conditions [82]. In these studies, the HP method effectively distinguished between the λ of different soil types, providing crucial data for environmental and agricultural applications. Another significant example involves the use of a distributed heat pulse sensor network for thermo-hydraulic monitoring, which emphasized the method's ability to capture temporal and spatial variations in soil thermal properties [83]. These cases underscore the HP method's adaptability and precision, making it a valuable tool for soil scientists requiring reliable data on soil thermal dynamics across diverse environmental conditions.

Despite its advantages, the HP method faces several equipment and methodological limitations that can affect its effectiveness in assessing soil λ (Appendix: Fig. 6). One of the primary challenges is the calibration of equipment, which requires precise adjustments to ensure accurate thermal property measurements [82]. Inadequate calibration can lead to errors in data analysis, as minor inconsistencies in sensor measurements may result in substantial inaccuracies in the calculated λ of soil. Additionally, the method's dependency on specific environmental conditions can limit its applicability, as factors such as soil variability and thermal gradients can introduce inconsistencies in results [78]. Consequently, careful consideration of these factors is crucial for obtaining reliable data, underscoring the need for continuous methodological refinements and technological advancements to improve the accuracy and applicability of the HP method in various settings.

Table 2b highlights the limitations of the HP method in measuring soil λ , particularly in terms of calibration, data analysis, and field applications. It also emphasizes the need for better integration of soil properties and a deeper

understanding of environmental factors that may influence measurements. Future research should focus on improving calibration techniques, developing advanced data analysis models, and conducting field studies to enhance the accuracy, reliability, and applicability of the HP method under diverse environmental conditions. By addressing these research gaps, the HP method can be further refined to improve its overall effectiveness and reliability in assessing soil thermal conductivity.

3.2.3 Transient Plane Source (TPS) Method

The Transient Plane Source (TPS) method is a widely used technique for assessing soil λ , which is crucial for various engineering applications, including geothermal energy systems and construction in permafrost regions. Also known as the Hot Disk method [39–41], this method is valued for its ability to measure λ and heat capacity with a declared uncertainty of approximately 5% [84]. Due to its sensitivity to small variations in λ , the TPS method is a reliable choice for detailed soil analysis. Consequently, its application enhances management and design strategies in heat-related soil studies, making it essential to explore its procedural intricacies and practical benefits in greater depth.

This technique is particularly advantageous in situations where soil samples are limited in volume [85], as it requires only a minimal quantity of soil to produce reliable findings (Appendix: Fig. 7). The rapid measurement capability of the TPS method enhances its practicality for field applications, allowing for timely assessments of λ in diverse soil types. As a result, the TPS method is integral to advancing research and applications in fields such as agriculture and environmental management, where soil thermal properties play a pivotal role in system performance and sustainability [86].

The procedure for employing the TPS method to measure soil λ involves several distinct steps to ensure accurate data collection (Appendix: Fig. 7). Initially, a TPS sensor, typically composed of a thin disk, is inserted into the soil sample, serving as both a heat source and a temperature recorder [85, 87]. Once the sensor is positioned, an electrical current is applied, producing thermal energy that propagates through the soil [40]. This heat transfer is monitored by the sensor, which records temperature changes over time, a crucial factor for calculating λ [86]. The recorded data are subsequently processed using established mathematical models, allowing for an accurate determination of soil thermal conductivity and facilitating a comprehensive understanding of soil thermal properties.

The calculation process of the TPS method involves a series of mathematical operations that facilitate the interpretation of λ data obtained from soil samples. After measuring

temperature changes over time, the TPS method applies a theoretical model to analyze the transient temperature response. This model is based on solving the heat conduction equation for the specific sensor configuration, enabling the determination of λ values. The heat transfer data recorded by the sensor are fitted to the model to extract the thermal properties of the soil. By integrating these calculations with experimental data, the TPS method provides a precise and reliable assessment of soil λ , essential for understanding heat transfer dynamics in various environmental and engineering contexts. The TPS method offers several advantages when assessing soil λ (Appendix: Fig. 7), notably its ability to provide rapid and accurate measurements with minimal sample preparation [85]. This efficiency is particularly valuable in practical applications, where time constraints are a concern and rapid data collection is necessary [88]. However, the accuracy and reliability of the TPS method can be influenced by several factors, including soil μ [87], sample homogeneity, and temperature stability during λ measurement [89–91]. Variations in μ can significantly impact λ readings, making it necessary to carefully control and monitor levels to ensure consistent results [92]. Additionally, the homogeneity of the soil sample is crucial, as heterogeneous samples may introduce errors in λ measurement, affecting the precision and reliability of the TPS method across different soil conditions. Despite its numerous advantages, the TPS method faces several limitations in measuring soil λ that must be considered. One major challenge is the potential for systematic errors, especially when measuring low soil λ , which can influence the accuracy of the results [86, 88]. Additionally, external environmental conditions, such as temperature fluctuations, can alter the soil thermal properties, leading to inconsistent λ measurements. Another limitation arises from the assumption of homogeneity in soil samples; in reality, soil often exhibits heterogeneity, which may affect the precision of TPS measurements. Addressing these challenges requires refinement of experimental setups and the advancement of calibration methodologies to improve the reliability and applicability of the TPS method across various soil environments. As presented in Table 2c, these challenges and opportunities for improving the TPS method are outlined, particularly in terms of soil heterogeneity, μ dynamics, and temperature range limitations. The table also highlights the need for better calibration standardization, expanded field application studies, and integration with advanced technologies. Future research should focus on improving accuracy, reliability, and applicability of the TPS method through advanced calibration protocols, broader temperature and moisture studies, and modern sensing technologies. By addressing these research gaps, the TPS method can be further optimized to enhance

its effectiveness in evaluating soil thermal properties across diverse environmental and engineering applications.

3.2.4 Transient Hot Wire (THW) Method

The Transient Hot Wire (THW) method is a widely used technique for measuring soil λ by analyzing the heat transfer from a heated wire embedded in the medium [53, 93]. The measurement process follows a series of precise steps to ensure accurate λ determination [38]. Initially, a fine wire is inserted into the soil and connected to a power source, ensuring a controlled and uniform heating effect [94, 95]. As the wire heats up, the resulting temperature rise is recorded by strategically positioned sensors that capture the thermal response of the soil [96, 97]. These real-time temperature readings allow for λ calculation based on Fourier's law and a mathematical model tailored to the specific soil conditions [93, 98]. Given the method's sensitivity to measurement conditions, careful calibration and control of variables such as current and voltage are necessary to minimize errors, underscoring the importance of meticulously designed experimental setup to ensure accurate λ results [96, 99, 100]. (Appendix: Fig. 8).

The accuracy of the THW method is significantly influenced by various soil properties (Appendix: Fig. 8), particularly μ [101, 102] and ρ_b [93, 103]. The effect of μ on λ measurements arises from the fact that water has a higher λ than air, leading to variability in readings if not properly accounted for [92, 103]. Similarly, soil ρ_b affects measurements, as denser soils typically exhibit higher λ , necessitating careful calibration to maintain precision. These factors require meticulous adjustments and calibration of the instrumentation to manage the inherent variability introduced by different soil conditions [95]. Understanding these influences is essential for optimizing the performance of the THW method, reinforcing the need to consider soil-specific attributes during the λ measurement process to ensure reliable and accurate results.

Soil μ plays a particularly crucial role in λ measurements using the THW method. This variability necessitates adjustments in the measurement process to accurately reflect these changes [92]. Additionally, ρ_b is another significant factor, as denser soils allow for better heat transfer, thereby affecting the readings obtained during λ measurement. To account for these variations, precise calibration of the equipment is essential, ensuring that changes in soil properties are accurately incorporated into λ calculations [95]. By implementing compensatory measures for diverse soil conditions, the THW method enhances measurement reliability and precision, making it applicable across various environmental settings.

One of the primary advantages of the THW method in measuring soil λ is its combination of precision and efficiency, setting it apart from alternative techniques. A key benefit is its ability to acquire data rapidly, allowing for real-time analysis while minimizing the time required for thermal equilibrium, which can often be a limiting factor in other measurement methods [104]. This efficiency is further enhanced by the method's high precision, as its design inherently reduces errors associated with moisture movement and external environmental influences, leading to more reliable results. Additionally, the THW method can be used in situ, providing a more accurate representation of natural soil thermal properties without requiring extensive specimen preparation or transportation, which can introduce variability [105]. Consequently, this method is favoured for applications that demand swift, accurate, and contextually relevant λ measurements.

To ensure accurate λ measurements, the THW method requires specific equipment, including a fine wire, a precise power source, and sensitive temperature sensors. The wire, often made of platinum due to its stability and conductivity, is a critical component that must be maintained under constant tension to prevent movement during heating [95]. Despite its precision, several challenges can arise in the THW method, including the need to ensure uniform heating and to account for heat losses to the surrounding environment, both of which can introduce errors [99]. Moreover, the setup is sensitive to external conditions, such as ambient temperature fluctuations, making controlled laboratory environments necessary to maintain reliability [105]. Another limitation is the complexity of calibration procedures, which demand meticulous attention to detail to mitigate the effects of equipment variability and environmental factors on the readings [92]. Despite these challenges, the THW method remains a preferred choice for researchers and practitioners seeking efficient and contextually accurate soil λ measurements. Table 2d outlines the research gaps and future directions associated with the THW method, focusing on soil heterogeneity, variations in moisture and temperature, and calibration procedures. The table further emphasizes the need for improved field applications and integration with advanced technologies for soil λ assessment. Future work should aim to enhance accuracy, reliability, and applicability of the THW method through standardized calibration protocols, expanded temperature and moisture studies, and the integration of modern sensing technologies. By addressing these areas, the method's effectiveness in both laboratory and field settings can be significantly improved, ensuring more precise and reliable soil λ measurements.

3.2.5 Laser Flash (LF) Method

The Laser Flash (LF) method is an advanced technique used to measure the λ of material, including soil λ . The method operates by delivering a brief, intense laser pulse to the surface of a specimen, inducing a thermal response that is monitored over time [106]. This approach enables precise thermal property determination by analyzing the temperature rise over time, making it a highly efficient technique for measuring λ [107] (Appendix: Fig. 9). Despite its effectiveness in measuring λ in various materials, there is a notable lack of research regarding its use in soil science and environmental studies. This gap in the literature presents both challenges and opportunities for researchers. The LF method's ability to provide rapid and precise measurements makes it particularly valuable for practical applications in soil thermal studies. Given the complexity and variability of soil ecosystems, employing a method that balances precision with adaptability is crucial for understanding thermal dynamics across different soil types.

Adapting the LF method for soil applications requires careful modifications due to the heterogeneous nature of soil, which consists of varied particle sizes and moisture content. Standard measurement procedures must be adjusted to ensure accuracy. One critical factor is the preparation of soil samples, which must be homogenized and compacted to minimize errors in thermal diffusivity computations [108]. Additionally, calibration of the apparatus is essential to accommodate different soil types, as their variable thermal characteristics can significantly affect the accuracy of λ measurements. To address these challenges, researchers have developed specialized protocols that modify laser pulse duration and energy levels, optimizing the interaction with soil samples and enhancing measurement precision.

The LF method offers several advantages in measuring soil λ , particularly in terms of high precision, rapid data acquisition, and non-destructive testing. Its methodology allows for the accurate determination of thermal properties by quickly analyzing the temperature response to a laser pulse, making it ideal for studies requiring fast and reliable results. The technique facilitates expedited evaluations while preserving the integrity of the soil samples throughout the process.

Additionally, the compact and efficient nature of LF equipment, comprising a laser source, temperature sensors, and computational tools, enables efficient data analysis [106, 109]. The method's high-speed measurements allow for multiple tests within a short period, supporting comprehensive soil analysis while maintaining sample integrity. Moreover, the non-invasive nature of the LF method ensures that the physical composition of soil remains unaltered, preserving its natural attributes for further study. These

characteristics make the LF method an ideal choice for soil science applications, where both precision and sample preservation are critical, enhancing its utility in environmental and agricultural research.

While the LF method presents numerous advantages, it also faces limitations and challenges in soil measurements. One primary challenge is soil heterogeneity, which makes it difficult to apply the laser pulse uniformly, leading to variability in thermal measurements. Additionally, the presence of moisture and organic matter in soil influences its thermal response, necessitating careful calibration and sample preparation to reduce inaccuracies in λ values. Another limitation is the requirement for precise experimental control, particularly in laser pulse duration and energy levels, which can be technically complex and costly to maintain consistently across diverse soil environments. Given these challenges, future research should focus on enhancing measurement accuracy and reliability by refining calibration procedures, developing water content control protocols, and integrating modern sensing technologies. By addressing these areas, the LF method will become more effective in both laboratory and field settings, facilitating more precise and reliable soil λ measurements (Table 2e).

3.2.6 Time-Domain Reflectometry (TDR) Method

The Time-Domain Reflectometry (TDR) method is a versatile and efficient approach for measuring soil λ and other soil properties related to the energy balance. By integrating TDR with heat-pulse techniques, this method provides a comprehensive analysis of soil thermal properties [110, 111], making it valuable in agronomy, environmental management, and soil science research.

TDR employs electromagnetic wave principles to evaluate soil properties, including λ [112]. A TDR probe, typically composed of parallel metal rods, is inserted into the soil to establish a transmission line [113, 114] (Appendix: Fig. 10). When a voltage pulse is transmitted through this line, the travel time required of the pulse as it moves along the probe and reflects back is recorded [115, 116].

This time interval is used to calculate the dielectric constant of the soil, which is directly linked to its thermal properties, including λ [114, 115]. By analyzing the waveform reflections, the TDR method provides a rapid and non-invasive means of assessing soil λ . Furthermore, a TDR probe can simultaneously measure λ , thermal diffusivity, and volumetric heat capacity by integrating heat-pulse technology [110, 113]. This capability allows for multifaceted soil analysis at a single location, improving efficiency in soil research [117].

Several factors influence the accuracy of TDR-based soil λ measurements (Appendix: Fig. 10). Soil μ plays a crucial

role as it directly affects the dielectric constant, which TDR uses to infer λ [110, 113, 118]. Additionally, soil temperature can alter the speed of electromagnetic wave propagation, potentially affecting TDR readings [113, 119].

Moreover, soil texture influences soil dielectric properties, thereby impacting measurement precision [115]. By understanding how these factors affect TDR measurements, researchers can improve data interpretation and refine methodologies for more accurate soil thermal property measurements.

The TDR method offers several advantages, making it an attractive choice for soil researchers and practitioners. One of the primary advantages is its non-invasive and repeatable nature, allowing for repeated λ measurements at the same location without modifying soil structure [115, 120]. This feature is especially beneficial in long-term environmental and agricultural studies, where maintaining soil integrity is crucial. Additionally, TDR provides rapid and efficient in data collection [29], facilitating quick evaluations, which is vital for dynamic environments. Furthermore, TDR measurements are known for their consistent and dependable results, even under changing environmental conditions, making them highly useful for scientific research and management practices [118, 121].

Despite its advantages, the TDR method presents certain limitations (Table 2f). Frequent calibration is required to maintain measurement accuracy, particularly in heterogeneous and changing soil environments [119]. The cost of TDR probes and related equipment can be a barrier to widespread adoption, especially in resource-limited settings [118]. Additionally, potential sources of error, such as electrical noise, probe misalignment, and environmental variability can introduce inaccuracies in λ measurements [122]. Other constraints affecting the utility of TDR include: (i) In high-salinity soils, the presence of salts affects electromagnetic wave interactions, impacting dielectric constant calculations and, consequently, λ estimates [115]; (ii) layered or stratified soils can cause inconsistent readings, as TDR assumes uniformity in the probed area [29, 118]; and (iii) variations in ambient temperature can introduce errors, necessitating additional calibration or compensation techniques to ensure data accuracy [29].

To enhance the accuracy, reliability, and applicability of the TDR method, future research should focus on: (i) Developing standardized calibration protocols to improve measurement consistency; (ii) implementing advanced moisture control techniques to mitigate the impact of soil water content variability; and (iii) creating continuous monitoring systems to enable long-term field applications. By addressing these challenges, the TDR method will become more effective for both laboratory and field applications, ensuring

precise soil λ measurements and supporting environmental and agricultural advancements (Table 2f).

3.3 Advanced Measurement Techniques

Advanced techniques for measuring soil λ have become increasingly crucial in environmental and agricultural sciences. Accurate λ measurements are essential for understanding heat transfer processes in soil, which influence crop growth, irrigation strategies, and climate modelling. Traditionally methods often relied on invasive techniques that disrupted soil properties, potentially affecting measurement accuracy. However, the development of non-invasive technologies such as fiber optic sensors (FOS) and wireless sensor networks (WSN) has introduced novel approaches for accurate and continuous λ monitoring (Fig. 1).

The FOS method employs distributed temperature sensing (DTS) to capture thermal variations along optical fibers embedded in the soil. This technique provides high-resolution spatial and temporal data while preserving soil integrity, making it particularly suitable for large-scale and long-term studies. Similarly, WSN technology utilizes sensor nodes to collect real-time soil temperature and moisture data, both of which influence λ . By enabling simultaneous measurements across different locations, WSNs enhance the comprehensiveness of soil thermal analysis while minimizing manual data collection.

These advanced techniques offer significant advantages over conventional methods. Their non-invasive nature ensures minimal soil disturbance, while their ability to provide real-time monitoring improves the accuracy and efficiency of λ assessments. The integration of FOS and WSN in soil research enhances decision-making in environmental management and agricultural applications, ensuring more reliable and continuous observations of soil thermal properties.

3.3.1 Fiber Optic Sensors (FOS) Method

The FOS method represents a significant advancement in the measurement of soil λ , offering technological improvements that enhance precision and efficiency [25, 123]. These sensors function by transmitting light through optical fibers [124], where variations in the light's properties indicate changes in soil temperature and μ [125]. This approach enables the acquisition of high-definition data, which is crucial for accurately characterizing soil thermal properties (Appendix: Fig. 11), it presents a schematic representation of the FOS methodology, detailing the measurement process, sensitivity to soil properties, advantages, and necessary instrumentation. A key advantage of the FOS method is its minimal invasiveness, reducing disturbance to the soil

ecosystem while preserving its natural conditions [126]. Additionally, the ability to provide continuous and real-time data allows for dynamic monitoring of soil conditions, leading to more informed decision-making in agricultural and environmental applications [25]. The working principle of FOS relies on detecting changes in soil temperature and moisture through light transmission within optical fibers [25, 125]. These fibers, composed of a core, cladding, and protective coating, facilitate precise light transmission, thereby enhancing λ measurement accuracy [127]. Environmental variations alter the light's characteristics, such as intensity and phase, which can be analyzed to determine soil thermal properties [126].

The measurement process using FOS method for soil λ begins with the careful installation of the optical fibers in the soil. These fibers are strategically positioned to maximize interaction with the soil's thermal properties, ensuring precise data collection. Once in place, a light signal is transmitted through the fibers, continuously monitoring temperature-induced variations in the light [123]. These fluctuations directly correspond to the soil's λ , providing real-time insights into its thermal dynamics [126]. The collected data is then processed to interpret λ values, offering a comprehensive assessment of soil thermal behaviour under various conditions and ensuring adaptability to different environmental settings [128].

The sensitivity of the FOS method to soil properties is notably influenced by variables such as μ and soil texture [25, 124]. Soil μ plays a crucial role, as it directly affects thermal properties by modifying heat capacity and λ , thereby influencing the accuracy of FOS measurements [125, 129]. Similarly, soil texture, which is determined by the distribution of sand, silt, and clay particles, impacts the interaction between optical fibers and soil components. This interaction can lead to variations in the light signal, potentially affecting the precision of λ measurements [126]. Consequently, understanding the heterogeneity of soil properties is critical for optimizing FOS deployment and ensuring accurate data collection across different soil types and environmental conditions.

The FOS method is particularly advantageous for soil analysis due to its high sensitivity and ability to provide continuous, real-time data, which are essential for dynamic environmental monitoring [130–132]. By minimizing soil disturbance through several effective strategies as specific plough design, waiting sometime after to restructure after cable installation, and conducting multiple saturation–drainage cycles, this method enables accurate long-term data collection without significantly altering the natural soil structure, making it highly effective for both agricultural and environmental applications [133–135].

Despite the numerous benefits, the FOS method also presents certain limitations, particularly in terms of equipment and methodology, as outlined in Table 3a. One primary challenge is the complexity of the initial setup, which requires precise installation of optical fibers to ensure reliable data collection. Additionally, data interpretation may be complicated by external factors such as temperature fluctuations and soil heterogeneity, potentially affecting measurement accuracy [136, 137]. The durability of optical fibers is another concern, as they may degrade over time or become damaged, necessitating careful handling and maintenance [126].

Furthermore, the high cost of FOS technology can limit its widespread adoption, particularly in resource-constrained settings, restricting its accessibility and implementation in broader agricultural and environmental applications [25, 138, 139]. Addressing these challenges is essential to maximizing the potential of FOS in soil thermal conductivity measurements. Future research should focus on improving calibration techniques, understanding soil and temperature effects, and overcoming field deployment challenges to enhance the accuracy, reliability, and applicability of the FOS method in diverse environmental conditions (Table 3a).

3.3.2 Wireless Sensor Networks (WSNs)

The Wireless Sensor Networks (WSNs) have emerged as a promising technology for measuring soil λ , a key parameter in understanding soil heat transfer processes (Abed Gatea [63; 140]). The WSNs method consists of spatially distributed sensors that collect and transmit data wirelessly, enabling continuous monitoring without the need for extensive manual intervention [141, 142]. The integration of WSNs in agri-environmental contexts not only enhances data precision but also facilitates real-time analysis of soil conditions [143, 144]. This discussion explores the role of WSNs in accurately measuring soil λ , emphasizing their potential to optimize agricultural practices and environmental monitoring. By examining their capabilities and challenges, we gain a deeper understanding of their importance in promoting precision agriculture and sustainable land management. The WSNs employ various sensor technologies and methodologies to capture soil λ , which is crucial for understanding energy and water exchanges in the soil. The measurement process involves several key steps and principles (Appendix: Fig. 12). Initially, deploying wireless sensor nodes across the area of interest is essential for comprehensive spatial coverage and data collection. These sensors assess soil thermal conditions and μ , both of which are critical for calculating λ [145]. Once installed, the sensors transmit data wirelessly to a central analytical unit, where

Table 3 Research gaps and future work for the advanced measurement techniques in measuring soil λ

Method ID	Problems	Description of research gaps	Future work
Table 1a			
a) Fiber Optic Sensors (FOS) method	Calibration techniques	Lack of standardized calibration protocols for FOS-based λ measurements across soil types and environments	Develop robust, standardized calibration procedures that account for diverse soil textures and environmental conditions
	Soil composition and temperature effect	Limited understanding of how soil constituents (clay, sand, OM) and temperature variations influence FOS accuracy	Investigate FOS response to varying soil compositions and temperature ranges under different agri-environmental conditions
	Field deployment challenges	Field performance affected by soil disturbance, tillage, μ fluctuations, and environmental variability	Develop predictive ML models trained on lab and field data; validate FOS performance under variable field conditions
	Cost-effectiveness and accessibility	High cost restricts broad adoption despite proven accuracy and efficiency	Test alternative sensor materials to reduce cost; evaluate performance and affordability in field trials
	Long-term monitoring	Limited studies on FOS durability and performance for continuous λ monitoring	Assess long-term stability and maintenance under diverse agricultural systems; use ML for real-time calibration optimization
	Integration with other λ techniques	FOS integration with other λ measurement methods (LF, THW, TDR) remains unexplored	Examine hybrid systems combining FOS with complementary methods to enhance accuracy and data reliability
Table 1b			
b) Wireless Sensor Networks (WSNs) method	Soil heterogeneity	Effects of soil heterogeneity on WSN λ measurements remain poorly understood	Examine how variations in soil texture and structure influence λ accuracy in WSN systems
	Calibration standards	Absence of standardized calibration protocols for WSN-based λ assessments	Develop and validate calibration standards using ML models to ensure accurate, cross-environment λ readings
	Temperature and μ interactions	Complex interactions between soil temperature and moisture affect λ estimation	Investigate joint effects of temperature and μ on λ to refine correction models for WSN data
	Network design optimization	Optimal sensor node configuration for diverse soils and conditions remains unclear	Optimize node placement, density, and communication range to enhance spatial accuracy and coverage
	Data management with field deployment challenges	Large data volumes and variable field conditions (tillage, climate) hinder reliability	Advance data analysis using ML; test WSN resilience under dynamic field conditions to improve stability and accuracy
	Long-term monitoring feasibility	Durability and maintenance requirements for long-term λ monitoring are underexplored	Evaluate WSN operational lifespan, reliability, and performance in extended agricultural applications

the information is processed to determine the soil's λ (Abed Gatea [63]). The automation of this process minimizes manual intervention while ensuring real-time data acquisition, thereby enhancing the precision and efficacy of λ measurements across diverse soil conditions [144].

Each sensor node within the WSN system is equipped to measure soil temperature [83] and μ [143], which are fundamental for assessing λ . These nodes communicate wirelessly with a central data collection unit, facilitating seamless data transmission and aggregation [146]. The collected data undergoes real-time analysis, enabling the rapid determination of soil λ , which is essential for timely decision-making in agri-environmental applications. By integrating sensors and advanced data collection techniques, WSNs allow for continuous monitoring [147], significantly improving the accuracy and efficiency of soil λ assessments [145, 148].

The implementation of WSNs in soil λ measurement offers several distinct advantages, primarily in real-time data collection and labour cost reduction. These networks facilitate continuous monitoring of soil parameters [149], providing immediate access to crucial data needed for informed decision-making in agronomic management [148]. The automation

of WSNs significantly reduces the need for manual data collection, thereby lowering labour expenses and minimizing human error [150]. Additionally, WSNs offer improved spatial and temporal resolution, allowing for high-accuracy λ data collection across extensive agri-environmental landscapes (Abed Gatea [63, 144]). This capability optimizes resource management and supports sustainable agricultural practices by enabling precise and timely interventions based on real-time soil condition assessments [83].

However, the accuracy of WSN-based soil λ measurements is highly sensitive to soil properties such as μ and texture (Appendix: Fig. 12). For instance, variations in μ can alter the soil λ , potentially affecting sensor readings [145]. Likewise, soil texture—determined by the relative proportions of sand, silt, and clay—modifies heat conduction pathways within the soil, influencing λ . These variations necessitate careful calibration of sensors to account for the specific soil conditions present at each measurement site [146]. Therefore, ensuring the adaptability of WSNs to diverse soil environments is critical for accurate and reliable data collection, underscoring the importance of addressing soil heterogeneity in λ assessments using WSN technology.

Despite their advantages, WSNs also present certain limitations related to equipment and methodological challenges, as summarized in Table 3b. These include issues such as soil heterogeneity, calibration difficulties, and temperature-moisture interactions. Signal disruption can also arise due to environmental factors, affecting data transmission between sensor nodes (Abed Gatea [63, 151]). Additionally, sensor durability is a concern, as exposure to μ , temperature fluctuations, and soil compaction may degrade sensor performance over time [146]. Methodologically, precise sensor calibration is essential but challenging, as it requires adjustments to account for varying soil properties at different sites.

Addressing these limitations requires the development of more resilient sensor materials and advanced calibration methodologies to improve the reliability and accuracy of WSNs in diverse agricultural and environmental settings [152]. Future research should focus on standardizing calibration protocols, optimizing sensor node configurations, and advancing data analysis techniques to enhance the overall efficacy of WSNs technology. These improvements will enable more effective and efficient monitoring of soil thermal properties, supporting both agricultural productivity and environmental sustainability.

4 Discussion and Limitations

This review explores the various measurement techniques used to assess soil λ and its significance in agricultural studies. Accurate λ quantification is essential for efficient soil management, influencing agricultural productivity under diverse environmental conditions.

Numerous techniques, both conventional and advanced, have been developed to assess soil λ , each with distinct advantages, limitations, challenges, research gaps, and future directions. Conventional techniques can be broadly classified into steady-state and transient strategies (Fig. 1), with each providing unique insights into soil thermal characteristics. The choice of technique depends on the specific application, required precision, and environmental conditions under which measurements are conducted.

Steady-state methodologies, including the Guarded Hot Plate (GHP), Thermal Cell (TC), Steady-State Needle Probe (SSNP), and Box Probe (BP), are recognized for their accuracy and reliability, particularly in laboratory settings. The GHP technique is highly effective in maintaining a uniform temperature gradient, essential for accurate λ measurements. However, it faces challenges, such as thermal losses and the impact of soil μ and composition on λ readings.

While the GHP method offers high precision, it has a notable design limitation, including the need for proper insulation and temperature control, which restricts its

applicability in field conditions. Additionally, it requires long testing durations, making it unsuitable for time-sensitive applications (Table 1a).

The TC technique provides high precision and the capability for in-situ λ evaluations, making it valuable for scientific and agricultural research. However, its costly and complex setup limits accessibility [51]. The insertion of TC probes into the soil may also disturb soil structure, affecting λ accuracy. Moreover, TC methods struggle with significant temperature gradients and require frequent calibration. Despite these drawbacks, TC techniques allow real-time data collection across different soil textures, making them essential for understanding soil thermal properties. The SSNP technique is particularly effective for measuring λ in unsaturated soils, as it accounts for dynamic pore moisture conditions. It is widely used in both laboratory and field studies due to its equipment requirements. However, improper probe insertion can cause measurement errors due to poor thermal contact with soil [27]. Additionally, thin probes may bend, leading to structural instability and further inaccuracies. Despite these challenges, SSNP remains valuable for agri-environmental investigations and engineering applications. Advances in probe design and data analysis techniques could enhance its precision and expand its applicability across diverse soil types (Table 1c).

The BP method is particularly beneficial for small sample analyses and field applications due to its minimal equipment requirements and relatively simple implementation. It has demonstrated good accuracy in measuring λ while maintaining reduced complexity. However, there is limited primary research on its application and optimization across various soil conditions and environments (Table 1d), which presents a gap in the current understanding of its full potential.

Several factors significantly influence the accuracy of BP measurements, including soil μ , soil composition, and temperature gradients. These variables can introduce variations in the results, necessitating further refinement of the methodology. Despite these challenges, studies have shown that the BP method aligns well with reference techniques such as the hot disk method, reinforcing its reliability. To improve its applicability, further research is required to optimize its performance under different environmental conditions. This is particularly crucial for field applications where traditional laboratory-based techniques may not be feasible or practical (Table 1d). Transient methodologies are highly regarded for their efficiency and ability to provide rapid and accurate measurements under diverse soil conditions. Some of the most widely used transient techniques include THIS, HP, TPS, THW, LF, and TDR methods. Each of these approaches has unique advantages and specific applications that contribute to a more comprehensive understanding of soil thermal properties.

Among these, the THIS method is commonly employed in both laboratory and field settings due to its ability to measure λ without requiring extensive soil sampling. However, its accuracy can be affected by soil μ levels and heterogeneity, which may lead to inconsistencies in measured values. This technique has proven to be particularly useful in agricultural applications, where precise assessments of soil thermal properties are essential for optimizing irrigation strategies and understanding the thermal dynamics that influence crop growth. Additionally, environmental scientists rely on this method to characterize soil thermal behaviour, which is crucial for assessing the impact of climate change on soil ecosystems.

Despite its widespread use, the THIS method still has limitations. Current calibration protocols do not fully account for variations in soil texture, structure and moisture content, which may introduce measurement discrepancies. Moreover, many existing setups require manual intervention, reducing their efficiency and making them less practical for large-scale or long-term studies. Addressing these shortcomings, Sakatani et al. [68] developed a compact THIS apparatus designed to measure the λ of extraterrestrial soils. The findings from their study indicated that the new apparatus yielded higher λ values than larger conventional setups. Numerical simulations confirmed that heat loss influences measurements, making it necessary to introduce calibration factors to improve the accuracy of λ assessment of soil samples. Future research should prioritize the development of standardized calibration protocols that account for soil variability while also exploring the integration of automated systems for data collection and analysis to enhance efficiency.

Another widely used transient method is the HP technique, which is recognized for its accuracy and non-invasive nature. This approach is particularly suitable for continuous monitoring in dynamic environments, making it a valuable tool for soil thermal studies. The HP method operates by introducing a heat pulse into the soil and monitoring the subsequent temperature fluctuations to determine thermal conductivity. However, the precision of this technique is highly dependent on several soil properties, particularly μ and texture. Soils with higher μ generally exhibit greater λ due to the enhanced heat transfer properties of water. Similarly, the texture of the soil, ranging from sandy to clayey compositions, significantly affects the thermal response, influencing the accuracy of the measurements [153].

Despite its advantages, the HP method also presents several challenges. Rigorous calibration is required to ensure accuracy, and errors can arise due to soil variability and thermal gradients (Table 1d). Zhao et al. [153] conducted a study utilizing the HP sensor to measure soil λ , identifying optimal heating parameters that minimize measurement

errors. Their research recommended specific ranges of accumulated heating energy and heating duration tailored for different soil types, which proved particularly useful for improving λ assessments in frozen soils. These findings highlight the importance of refining experimental parameters to enhance the accuracy and reliability of transient thermal conductivity measurements.

The TPS method is highly sensitive to minor variations in λ and is particularly useful for analyzing limited soil samples. This method involves inserting a TPS sensor into the soil, where it serves both as a thermal source and a temperature recorder. Its rapid measurement capability enhances its practicality for field applications, enabling timely assessments across diverse soil types. However, the accuracy of TPS measurements can be affected by factors such as soil μ , sample homogeneity, and temperature stability during λ assessment. Variations in μ can significantly alter readings, necessitating careful monitoring to ensure consistency. Additionally, the assumption of uniform soil composition does not always hold true in real-world conditions, leading to potential errors in λ determination.

Another transient technique, the THW method, is known for its rapid data acquisition and high precision. This approach involves embedding a fine wire within the soil and measuring the temperature rise as the wire is heated. The real-time analysis capabilities of the THW method reduce the time required for thermal equilibrium, making it particularly advantageous for studies requiring quick and efficient thermal measurements. However, this technique demands meticulous calibration and control of electrical variables such as current and voltage to minimize errors and ensure precise λ measurements. The accuracy of the THW readings is heavily significantly influenced by soil properties, particularly μ and pb . An increase in μ generally enhances λ due to water's superior conductivity compared to air. Likewise, denser soils tend to exhibit higher λ values, highlighting the necessity of precise calibration for reliable results.

Beyond conventional transient methods, advanced techniques such as FOS and WSNs represent significant advancements in λ measurement. FOS technology provides continuous, real-time data with minimal soil disturbance, making it particularly suitable for long-term environmental monitoring [25]. These sensors function by transmitting light through optical fibers, where variations in the light properties correspond to changes in soil temperature and moisture [137]. The adoption of FOS methods in soil analysis significantly reduces data collection intrusiveness, thereby preserving the natural condition of the soil ecosystem. Liu et al. [25] implemented FOS techniques, particularly the DPHP-FBG method, for estimating soil μ and λ , using a newly developed sensor. Both laboratory and field tests demonstrated the effectiveness of the DPHP-FBG system, which

offers advantages such as automated and wireless monitoring –particularly beneficial for remote areas and large-scale applications. However, the complexity of the initial setup and the substantial financial investment required for equipment may limit widespread adoption (Table 3a). Additionally, external factors such as temperature fluctuations and soil heterogeneity can complicate data interpretation, potentially impacting measurement accuracy [137].

Building on these advancements, Lin et al. [137] introduced a heated optical cable as a FOS tool to measure the λ of biocemented sand. Their study explored the use of microbially induced calcium carbonate precipitation (MICP) to enhance soil λ , particularly in arid environments. The FOS effectively detected thermal changes, providing insights that could optimize MICP strategies. Its built-in temperature field and high spatial resolution enabled continuous and distributed thermal response monitoring, offering a promising approach for improving soil thermal properties in challenging conditions.

WSNs present another innovative approach to λ measurements, offering real-time data collection with reduced labour costs, making them highly effective for large-scale agricultural and environmental applications. These networks consist of spatially distributed sensors that wirelessly collect and transmit data, enabling continuous monitoring without extensive manual intervention. The integration of WSNs into agri-environmental applications enhances data precision while facilitating instantaneous soil conditions analysis. Abed Gatea Al-Shammary et al. [63] demonstrated the use of distributed sensors for wireless data transmission, employing a 315 MHz RF Transmitter and Receiver Module to transfer measurement data from an Arduino board to a PC. This setup enabled remote data collection and analysis, significantly improving the efficiency of real-time thermal monitoring.

Despite their advantages, WSNs exhibit heightened sensitivity to soil properties such as μ levels and texture, which can influence the accuracy of λ measurements (Tables 3b). Additionally, challenges related to signal disruption and sensor durability under harsh environmental conditions remain areas of concern. [143] highlighted these limitations, noting that ongoing innovations aim to enhance WSN adaptability and precision in measuring soil thermal properties (Table 3b). While these issues persist, continuous advancements in sensor technology and data processing techniques hold great promise for improving WSN reliability in large-scale agricultural and environmental applications.

5 Summary of Knowledge Gaps and Future Research Directions

Steady-state methods, including the GHP, TC, SSNP, and BP, are recognized for their accuracy in controlled laboratory settings but face challenges such as heat loss, complex temperature control, and sensitivity to soil properties like moisture and texture. Future research should focus on improving insulation and temperature control systems for GHP, developing cost-effective and non-invasive TC probes, enhancing SSNP probe design for better soil contact, and validating BP methods in diverse field conditions. Additionally, the impact of soil composition and environmental factors on λ measurements requires further investigation to improve the reliability of these λ methods.

Furthermore, an important limitation of steady-state approaches lies in their restriction to point-scale measurements, which constrains their representativeness across heterogeneous agricultural landscapes. Future studies should explore strategies to upscale steady-state λ data by integrating them with remote sensing observations (e.g., thermal infrared or microwave data) to extrapolate local measurements to field and regional scales [32, 77]. Such multi-scale integration would enhance the spatial applicability of these methods and enable improved model calibration in large-scale soil–climate interactions.

Transient methods, including the THIS, HP, TPS, and THW, offer rapid and efficient λ measurements but are sensitive to soil moisture, inhomogeneity, and temperature gradients. Future research should aim to improve probe designs for better soil contact, automate calibration processes to reduce errors and refine data analysis techniques to account for soil heterogeneity. The development of distributed sensor networks for HP methods and the application of TPS in diverse soil environments are also promising directions. Additionally, studying the impact of soil properties on transient measurements will enhance their accuracy and applicability.

Another key direction involves linking transient measurements with satellite-derived surface temperature products or thermal inertia indices to validate and refine remote sensing–based estimates of soil thermal behavior [154, 155]. By combining in-situ transient data with Earth observation datasets, it becomes possible to build predictive models that connect localized λ measurements with regional soil thermal regimes, bridging the gap between point-scale observations and remote sensing spatial coverage.

Advanced methodologies, including FOS and WSNs, provide continuous, real-time λ assessments with minimal soil disturbance. However, their high costs, complex setups, and sensitivity to environmental variables limit their extensive application. Future research should focus on developing

cost-effective FOS systems, improving data interpretation techniques, and enhancing the durability of WSN sensors in adverse conditions. Integrating these advanced techniques with additional soil property sensors can provide a comprehensive understanding of soil health and support precision agriculture.

The spatial continuity and real-time data provided by FOS and WSN networks also offer significant potential for multi-scale integration. When combined with airborne or satellite-based remote sensing, these networks can act as validation nodes to ensure that remotely sensed surface thermal patterns accurately reflect subsurface heat transfer processes [156, 157]. Such integration is crucial for developing high-resolution, spatially explicit maps of soil thermal conductivity and for enhancing digital soil mapping frameworks.

Artificial intelligence (AI) and machine learning (ML) offer promising avenues for enhancing data analysis methodologies, predictive modelling, and calibration procedures for soil λ measurements. Standardizing protocols for calibration and measurement across different methods and soil types will improve the reliability and comparability of results. Research should also examine the effects of climate change on soil λ and its implications for farming practices. Furthermore, the development of high-resolution soil λ maps for precision agriculture and interdisciplinary collaboration among soil scientists, agronomists, and engineers will advance the understanding of soil thermal dynamics and promote sustainable land management.

Ultimately, integrating field-scale λ data (obtained from steady-state, transient, and advanced methods) with remote sensing and AI-based modelling will allow for scalable predictions of soil thermal conductivity across heterogeneous landscapes. This approach bridges the gap between localized measurements and regional or global soil thermal assessments, enabling more accurate parameterization of land surface models and contributing to climate adaptation research [155, 158].

6 Conclusions

Numerous methods have been developed to measure soil λ , each with distinct advantages and limitations. Steady-state techniques, such as GHP and TC, are widely recognized for their precision in controlled laboratory environments. However, these methods often require long testing durations and are sensitive to soil properties such as soil water content and texture. In contrast, transient methods, including HP and TPS, enable rapid and efficient measurements but can be affected by soil inhomogeneity and environmental variability. Advanced techniques, such as FOS and WSNs, facilitate

real-time, non-invasive monitoring. However, their widespread application is often limited by high costs and complex setups.

Despite advancements in soil λ measurement techniques, significant knowledge gaps remain. Key challenges include the need for improved calibration methods, a deeper understanding of how soil properties affect λ measurements, and the development of cost-effective and portable measurement devices. Future research should focus on integrating advanced technologies, such as artificial intelligence and machine learning, to enhance data analysis, predictive modelling, and calibration accuracy. Additionally, the establishment of standardized measurement protocols and further exploration of climate change impacts on soil thermal dynamics are critical. Interdisciplinary collaboration among soil scientists, agronomists, and engineers, along with the development of high-resolution soil λ maps, will enhance precision agriculture and support sustainable land management practices.

In conclusion, while significant progress has been made in the study and quantification of soil λ , continued research and technological advancements are essential to address existing challenges. By improving measurement accuracy, efficiency, and accessibility, researchers can contribute to the development of more sustainable agronomic practices, enhance soil health, and optimize agri-environmental management.

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Declarations

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